



Crack Width Calculator

User Guide | EN 1992-1-1 cl. 7.3.4

This app calculates the characteristic crack width w_k of a reinforced concrete section and checks it against a limit. It is valid for slabs only - not beams or columns: the check assumes a slab-type section (a rectangular strip with one bar layer per face), so beam or column dimensions return a crack width that does not apply to them.

1. Launching the app

Double-click CrackWidthNiceGUI.exe. It opens in its own window; no installation is needed. Keep the whole folder (with _internal) together.

2. Step-by-step workflow

1. Fill in the four input panels: **Geometry**, **Reinforcement**, **Materials**, **Loads**. Click a panel header to expand it.


Crack Width Calculator (Wiecon)
CREDITS

Inputs

 **REFERENCE IMAGE**

 **Geometry**

Section width b [mm]
1000.00

Section thickness h [mm]
525.00

Cover to bar surface c [mm]
40.00

 **Reinforcement**

Tension face bar $\varnothing$ [mm]
16.00

Tension face bar spacing [mm]
150.00

Opposite face bar $\varnothing$ [mm]
20.00

Opposite face bar spacing [mm]
150.00

Bar type
ribbed

 **Materials**

Concrete strength f_{ck} [MPa]
50.00

Concrete modulus $E_{c,c}$ [GPa] (0 = auto)
34.00

Steel modulus E_s [GPa]
200.00

Creep coefficient φ
0.00

 **Loads**

Axial force N [kN] (tension +)
529.00

Bending moment M [kNm]
116.00

Crack width limit w_{max} [mm]
0.30

Load duration
long

Summary of inputs

GEOMETRY

Section width b [mm] **1,000.00**

Section thickness h [mm] **525.00**

Cover to bar surface c [mm] **40.00**

REINFORCEMENT

Tension face bar $\varnothing$ [mm] **16.00**

Tension face bar spacing [mm] **150.00**

Opposite face bar $\varnothing$ [mm] **20.00**

Opposite face bar spacing [mm] **150.00**

Bar type **ribbed**

MATERIALS

Concrete strength f_{ck} [MPa] **50.00**

Concrete modulus $E_{c,c}$ [GPa] (0 = auto) **34.00**

Steel modulus E_s [GPa] **200.00**

Creep coefficient φ **0.00**

LOADS

Axial force N [kN] (tension +) **529.00**

Bending moment M [kNm] **116.00**

Crack width limit w_{max} [mm] **0.30**

Load duration **long**

Fig 1 — The four input panels; a live summary of inputs sits below them.

2. Review the "Summary of inputs" below the panels to confirm your values.

3. Watch the "Result" badge. It recalculates as you type, so there is no Calculate button. It reads $w_k = 0.288$ mm $\leq w_{max} = 0.300$ mm (Okay) on green (within the limit), or $w_k = 0.678$ mm $> w_{max} = 0.300$ mm (Not Okay) on red (exceeds it), and shows a grey dash while any field is still empty or half-typed.

 **Result**

 $w_k = 0.288$ mm $\leq w_{max} = 0.300$ mm (Okay)

 **Save File**

Fig 2a — Within the limit: the badge turns green and reads (Okay).

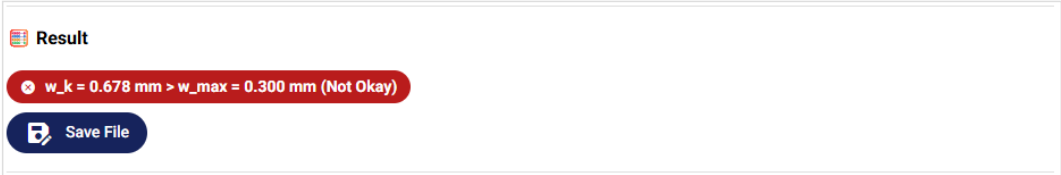
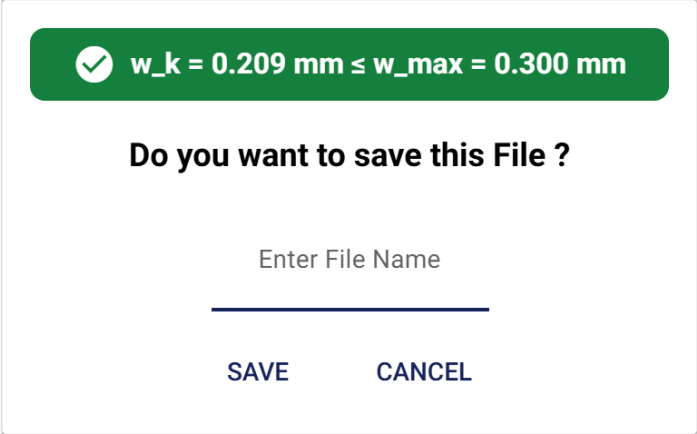
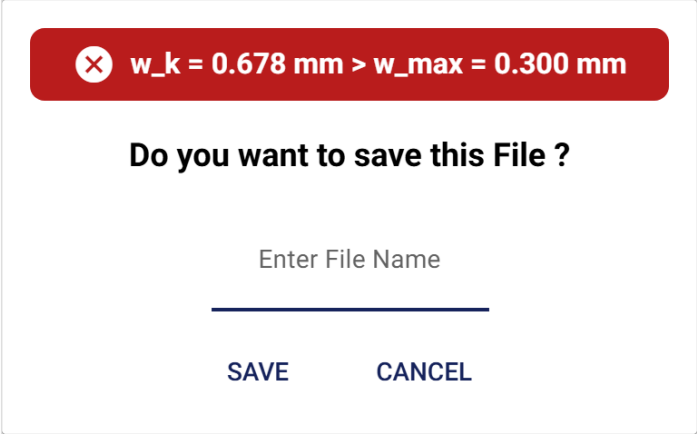


Fig 2b — Over the limit: the badge turns red and reads (Not Okay).

4. Save the result: click **Save File**, type a **File name**, then click **Save** (or press **Enter**). The dialog repeats the verdict so you can confirm first.



PASS — the save dialog repeats the verdict



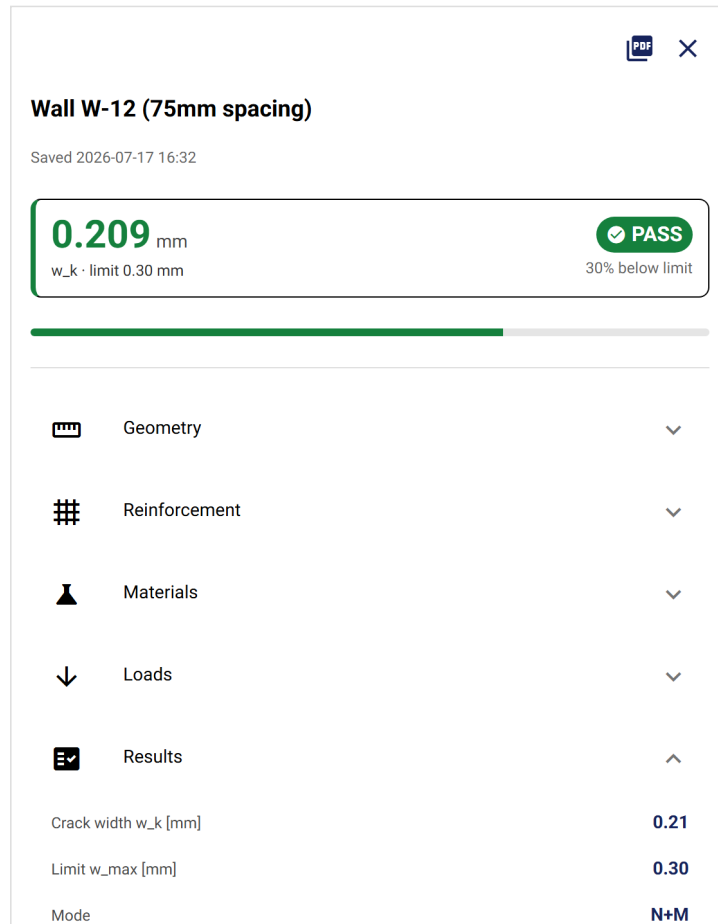
FAIL — the save dialog repeats the verdict

5. Manage saved files in the **Saved Files** table: **View More Detail**, **Delete File**, or **Delete All Files**.

☐	File Name	Created at	Crack width w...	Limit w_max [...]	Pass	Mode	Steel stress σ...	Crack spacing...
☐	Wall W-12 (75m...	Jul 17, 2026 04:3...	0.209	0.30		N+M	198.6	350
☐	Wall W-12 (150...	Jul 17, 2026 04:3...	0.678	0.30		N+M	395.7	565

Fig 3 — Saved files, colour-coded by verdict: PASS (green), FAIL (red).

6. Export to PDF: in the detail view, click the **PDF icon** (one page per file). PDFs go to your **Downloads** folder.



Wall W-12 (75mm spacing)

Saved 2026-07-17 16:32

0.209 mm
w_k · limit 0.30 mm

PASS
30% below limit

- Geometry
- Reinforcement
- Materials
- Loads
- Results

Crack width w _k [mm]	0.21
Limit w _{max} [mm]	0.30
Mode	N+M

Fig 4 — Detail view; the PDF icon (top-right) exports it as a report.

3. Importing multiple cases at once

To run a batch of cases without typing each one, use the Import and Download button (beside Reference Image). It expands to two actions: download a blank template, and upload a filled-in file.

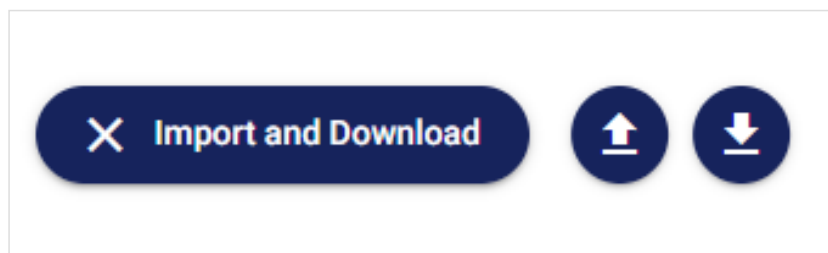


Fig 5 — The Import and Download button expands to a template download and an upload action.

Download the template first if you are unsure of the format: it is a .csv with one column per input field. Enter one case per row, then save the file.

Upload the file: click the upload action and choose your .csv. Each row is calculated and stored as a saved case, added to the Saved Files table; a progress bar tracks the import and a message confirms how many were added. Only .csv files are accepted.

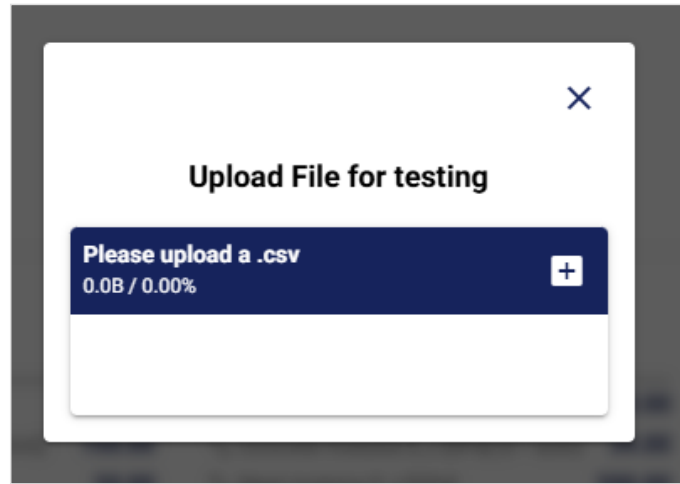


Fig 6 — The upload dialog; browse to a .csv to import its rows.

4. Handy controls

Dark-mode toggle (the round button, bottom-right) switches the whole app between the light and dark themes. It shows a sun while you are in light mode and a moon while you are in dark mode — the icon is the mode you are in, not the one you will get. The choice applies everywhere: inputs, summary, result badge and the Saved Files table.

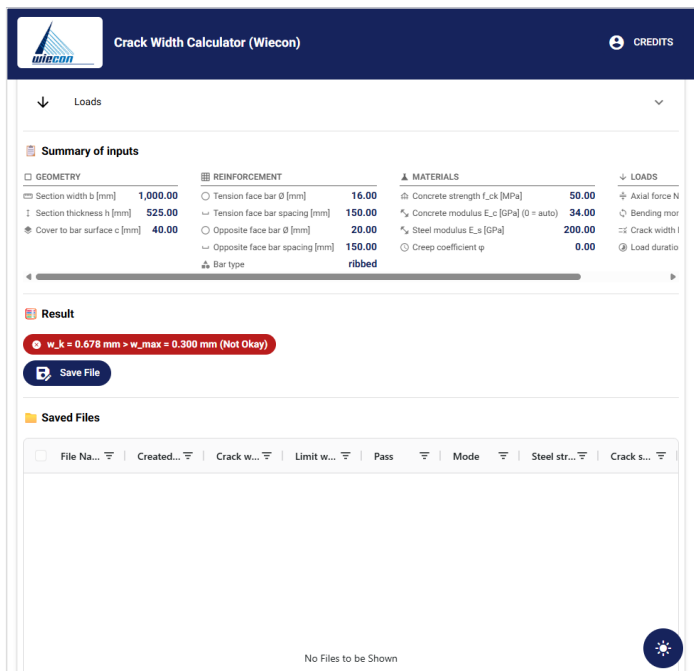


Fig 5a — Light mode (the default)

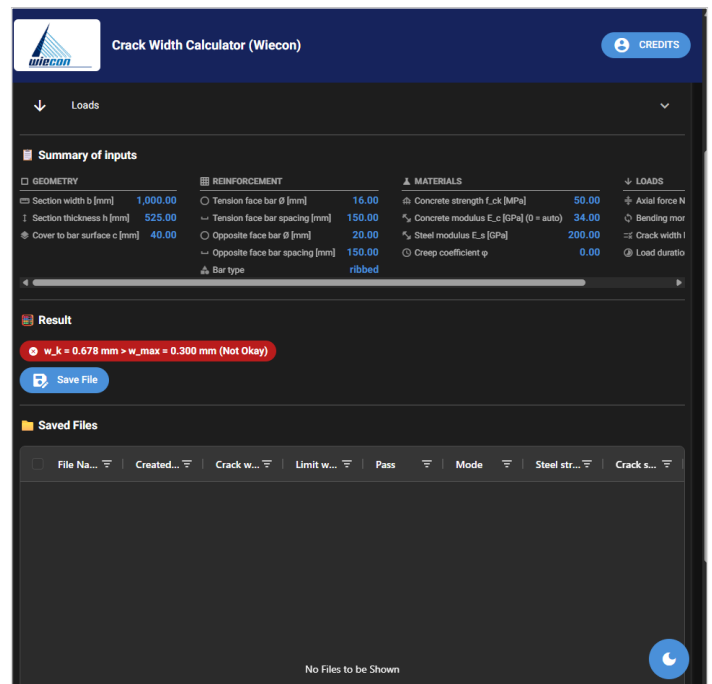


Fig 5b — Dark mode

Press 'f' to toggle fullscreen. The Credits button in the header shows the app info and the standard reference.

The Help button (next to Credits) opens a "How to use this tool" guide: a numbered walkthrough with a screenshot for each step. Click the zoom icon on any screenshot to view it full-size, or Watch walkthrough to play a short video that can be opened full-screen.

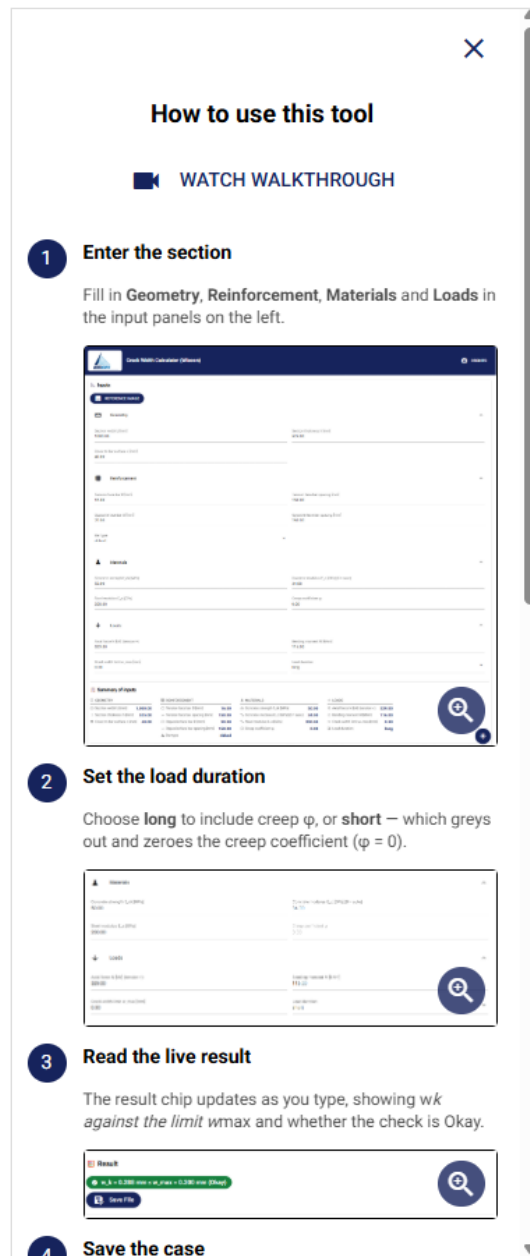


Fig 7 — The Help guide: a numbered walkthrough with a zoomable screenshot for each step.

Right-click the header logo for a quick menu: Toggle Light/Dark Mode (the same switch as the round button), and Exit to close the application.

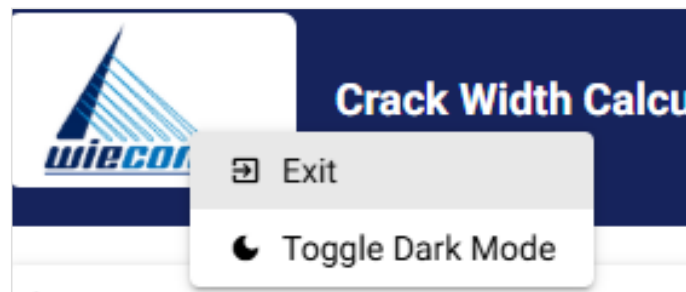


Fig 8 — Right-click the logo for Exit and the theme toggle.

5. Input field reference

Geometry

Field	Meaning	Unit	Default
Section width b	Width of the section	mm	1000
Section thickness h	Overall thickness	mm	525
Cover to bar surface c	Clear cover to bar surface	mm	40

Reinforcement

Field	Meaning	Unit	Default
Tension face bar Ø	Bar diameter, tension face	mm	16
Tension face bar spacing	Centre-to-centre spacing	mm	150
Opposite face bar Ø	Bar diameter, opposite face	mm	20
Opposite face bar spacing	Centre-to-centre spacing	mm	150
Bar type	ribbed (high bond) or plain	-	ribbed

Materials

Field	Meaning	Unit	Default
Concrete strength f_{ck}	Characteristic cylinder strength	MPa	50
Concrete modulus E_c	Elastic modulus; 0 = auto	GPa	34
Steel modulus E_s	Elastic modulus of steel	GPa	200
Creep coefficient ϕ	Creep coefficient, long-term	-	0

Loads

Field	Meaning	Unit	Default
Axial force N	Axial force; tension positive (+)	kN	529
Bending moment M	Applied bending moment	kNm	116
Crack width limit w_{max}	Allowable crack width	mm	0.30
Load duration	long (sustained) or short	-	long

Note: a blank field reads as empty and the Result badge simply shows a dash until every field is filled; if you click Save File with one missing, the app names the value it needs. The badge updates shortly after you stop typing, not on every keystroke, so a half-typed number never flashes a misleading verdict. Results are stored locally and reload next time you open the app.